

31 JULY 2026

TECHNO-COMMERCIAL OFFER

For 1.5 MLD STP

with an Advanced MBR System



K.V.G. Medical College & Hospital



ETP at CAMPCO, Puttur



Shed No. C-54, Behind Bank of Baroda, Baikampady Industrial Area, Mangalore

Ref. No.: QT/STP/2026-27/76

Date: July 31st, 2026

Offer to: **K.V.G. Medical College & Hospital**
 Court Rd, Kurunjibhag, Sullia, Karnataka
 574327

Kind Attn.: Mr. Chidanand

Basic value: 2,90,00,000.00

Valid upto: 6 Weeks from date of Ref

1. INTRODUCTION:

About Aapaavani Environmental Solutions Pvt. Ltd.

Aapaavani is an environmental consultancy based in Mangalore, Karnataka founded by qualified environmental engineers. Equipped with technical competence and experience in the field of environmental engineering and environmental legal compliance, Aapaavani strives to help all kind of industries, societies, communities, apartments in fulfilling environmental compliance with best available technology (BATs), economic and feasible solutions with customized as per the customer needs. Although the company is about 5+ years old, Aapaavani has taken up more than 75 projects in the short span including Sewage Treatment Plants, Effluent Treatment Plants, Water Treatment Plants, Hydrogeological surveys, Environmental clearances, compliance documents etc.

In future, Aapaavani Environmental Solutions Pvt. Ltd. will be referred as “Supplier”

Meet the Team



Mr. S Somashekar
VP-Operations



**Dr. Vishnu Sharma
Atikukke**
Co-Founder & CTO



**Mr. Chinmaya
Delampady**
Co-Founder & CEO



Mr. B Krishnappa
Senior Consultant
Environment



Dr. H N Udaya Shankar
Senior Consultant
Hydrogeologist

PROMINENT CLIENTS:

EFFLUENT TREATMENT PLANTS



SEWAGE TREATMENT PLANTS



SERVICE STATION EFFLUENT TREATMENT PLANTS



DETAILED PROJECT REPORT



About K.V.G. Medical College & Hospital:

K.V.G. Medical College & Hospital, Sullia, established in 2002, is a renowned medical institution under the K.V.G. Educational Trust, committed to providing quality medical education, advanced healthcare services, and clinical research. Affiliated with the Rajiv Gandhi University of Health Sciences and recognized by the National Medical Commission, the institution offers undergraduate and postgraduate programs in medicine and allied health sciences, supported by experienced faculty, modern teaching methodologies, and extensive clinical exposure. The college emphasizes academic excellence, ethical medical practice, research, and community healthcare, preparing students to become competent and compassionate healthcare professionals. The campus at Court Road, Kurunjibhag, Sullia, Karnataka, is equipped with modern lecture theatres, advanced laboratories, a well-stocked central library, skill laboratories, seminar halls, hostels, and other essential academic facilities. The attached multi-specialty teaching hospital provides comprehensive diagnostic, medical, surgical, emergency, and critical care services, offering students valuable hands-on clinical training while serving the healthcare needs of the region. Through continuous medical education programs, research initiatives, outreach activities, and patient-centered care, the institution promotes holistic development and professional excellence. With its strong commitment to quality medical education, healthcare delivery, and social responsibility, K.V.G. Medical College & Hospital has established itself as one of the leading medical institutions in Karnataka, producing healthcare professionals who contribute significantly to patient care, research, academia, and public health.

In future, K.V.G. Medical College & Hospital will be referred to as "**Client.**"

Design Parameters

Nature of Wastewater	Sewage
Flow	1500 m ³ /day
Operating Period	24 Hours

The plant is designed to treat generated effluent having following characteristics:

Characteristic of Waste and Treated Water		
Parameters	Waste Water	Treated Water
pH	6.5 – 8.0	6.5 – 8.0
COD	< 500 – 600 ppm	< 50 ppm
BOD (3 days at 27°C)	< 250 – 350 ppm	< 10 ppm
TSS	< 200 ppm	< 10 ppm
Oil & Grease	< 50 ppm	< 10 ppm

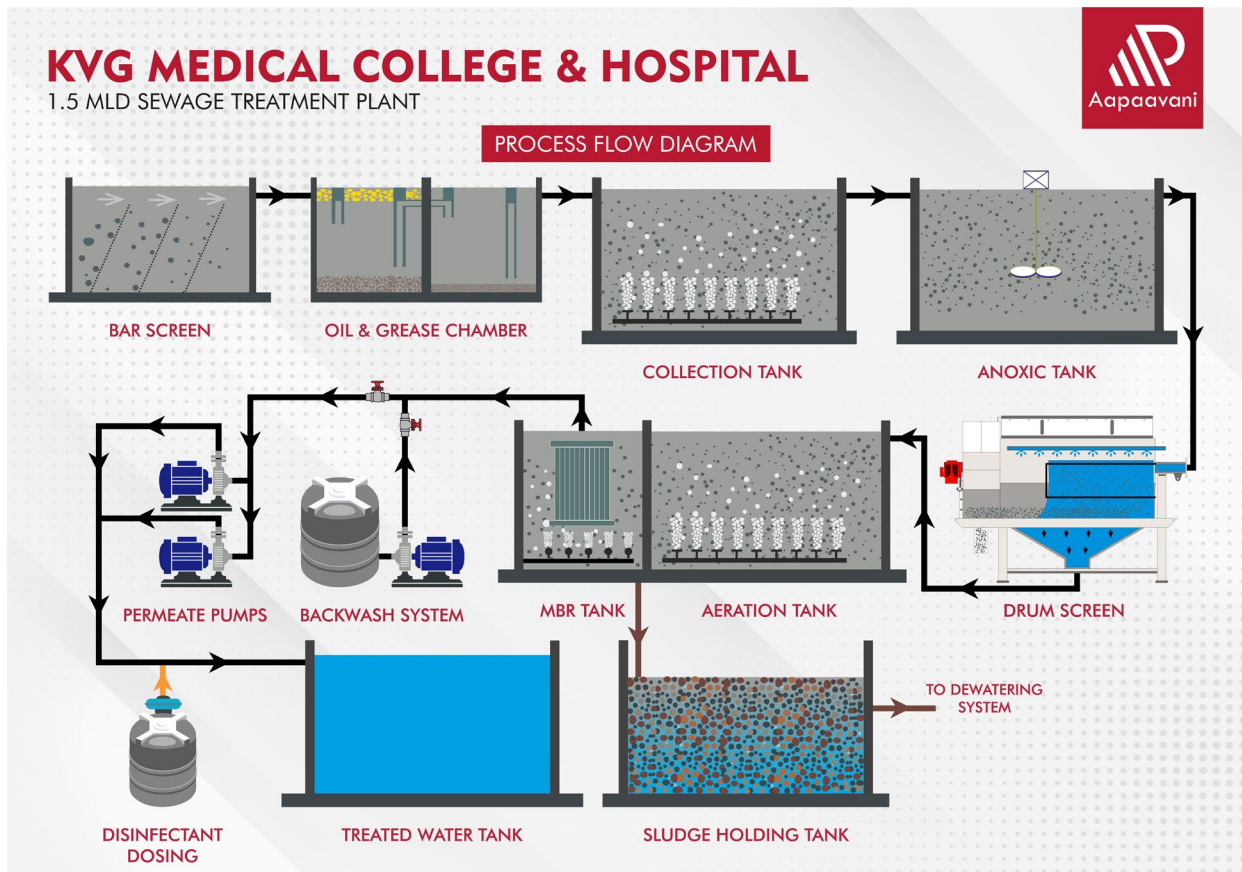
Assumptions

- The plant is designed to operate at +/- 10 % variation in wastewater parameter.
- No other parameters other than above is present in the wastewater which is beyond Pollution Control Norms and hazardous to micro-organisms.
- The expected treated water quality is achievable only if the raw water meets the specified parameters and contains no additional or excessive pollutants that could affect the biological process.

2. BRIEF PROCESS DESCRIPTION:

Aapaavani proposes a 1.5 MLD Sewage Treatment Plant (STP) based on advanced Membrane Bioreactor (MBR) technology for the efficient treatment of domestic sewage. The treatment process begins with a Bar Screen to remove coarse solids, followed by an Oil & Grease Chamber for the separation of oil, grease, and floating scum. The wastewater is then collected in the Collection Tank, which provides flow equalization before entering the Anoxic Tank for denitrification. It subsequently passes through the Drum Screen to remove fine suspended solids before flowing into the Aeration Tank, where aerobic microorganisms degrade organic pollutants. The mixed liquor is then treated in the MBR Tank, where submerged membrane modules produce high-quality treated water by retaining biomass and suspended solids. The permeate is pumped through Permeate Pumps, while the Backwash System periodically cleans the membranes to maintain filtration efficiency. After Disinfectant Dosing, the treated water is stored in the Treated Water Tank for reuse or safe discharge. Excess sludge is transferred to the Sludge Holding Tank and sent to the Dewatering System for volume reduction. The advanced MBR process provides a compact footprint, superior effluent quality, and reliable compliance with discharge and reuse standards.

Process flow diagram



Commercial Offer:

SL No	System - Type	Price in Rs.
MEMBRANE BIO REACTOR (MBR) TECHNOLOGY		
1	Sewage Treatment Plant – 1.5 MLD <i>Includes Supply, Installation and Commissioning</i>	2,90,00,000.00 (Rupees Two Crore Ninety Lakhs Only)

Civil in client scope

Total amount excludes 18% GST

Any changes in the tax structure by Government or local bodies, the difference has to be borne by client.

Payment Terms

- 60% advance payment along with the purchase order.
- 20% before dispatch of materials, against Performa invoice.
- 20% after installation and Commissioning.

Delivery period

- The delivery period for the entire plant shall be 2 months from the date of Purchase Order (PO), after the client has fulfilled all related responsibilities.

Equipment details:

1500 KLD STP ELECTROMECHANICAL ESTIMATION				
ROTARY DRUM SCREEN				
1	A Rotary Drum Screen is an automatic mechanical screening unit designed to remove coarse and fine suspended solids, fibrous materials and floating debris from the influent stream. Wastewater passes through a rotating perforated drum where solids are retained on the screen surface while the filtered water flows through. The retained solids are continuously removed by internal spray nozzles and discharged into a collection bin, ensuring efficient screening with minimal manual intervention and low maintenance.			
	Rotary Drum Screen	Design Flow: 100 m ³ /hr, Screen Opening: 2–4 mm MOC: SS 304	1	NO.
AERATION SECTION				
2	BLOWER & MOTOR			
	The blower and motor provide the necessary airflow to the air scoring system, facilitating efficient cleaning of MBR membranes. The blower and GI pipes are connected using flanges. A: Aeration tank and Sludge holding tank B: MBR Scouring and Collection tank			
	BLOWER with bed for A	FLOW 1200 m ³ /hr with 0.6kg/cm ² MAKE: Akash/equivalent	1 + 1	NO.
	MOTOR with bed for A	40 hp with 1440 RPM Make: CG/equivalent	1 + 1	NO.
	BLOWER with bed for B	FLOW 1200 m ³ /hr with 0.6kg/cm ² MAKE: Akash/equivalent	1 + 1	NO.
	MOTOR with bed for B	40 hp with 1440 RPM Make: CG/equivalent	1 + 1	NO.
3	BLOWER ACCESSORIES AND AIR DISTRIBUTION LINE: GI C class pipe			
	The blower accessories, such as filters, silencers, and control valves, along with the air distribution line, are essential components for the efficient operation of the blower system in the MBR process.			
	Main header, Distribution line, Diffuser connector	8-inch, 4 inch, 2.5 inches (Vary as per design). MOC: UPVC/GI	1	LS

4	ACCESSORIES AND FITTINGS FOR DISTRIBUTION LINE			
	Fittings and accessories for air distribution line.	Varying sizes of Butterfly valve, Flanges, NRV, Pressure gauge, Safety air vent, GI/MS fittings (TEE, Elbow, dummy, etc), Paints, Bolts and Nuts, etc, MOC: CI/GI/MS,	1	LS
	Fine bubble diffuser	Air flow rate: 5-6 m3/hr, Type: Fine bubble Disc type, size: 12 inch, MOC: PVC & EPDM rubber, MAKE: WTE Italy	90	NO.
	Coarse bubble diffuser	Air flow rate: 10-15 m3/hr, Type: Fine bubble Disc type, size: 5inch, MOC: PVC & EPDM rubber, MAKE: WTE Italy	38	NO.
MEMBRANE SECTION				
5	MEMBRANE SKID			
	Houses and supports the membrane modules in the MBR process. The material of construction is made of SS 304.			
	Membrane Skid with fabrication charges	The skid is fabricated as per design to meet the required holding capacity and suction requirements for the membrane modules. MOC: SS 304, MAKE: AES.	1	LS
	Guiding rod with fabrication.	For placement and easy movement of skid during CIP. MOC: SS 304, MAKE: AES	1	LS
6	Submerged hollow fiber membranes with Reinforced PVDF fiber. Each module has a membrane area of 40 sqm.			
	Membrane module as per design	Each module having 40 m2 surface area with 0.05-micron pore size. MOC: Reinforced PVDF fiber, MAKE: Mitsubishi	As per design	
	MEMBRANE LIFTING			
	For Membrane lifting along with manual monorail - sufficient for lifting the membrane in case of any cleaning of CIP as per SOP.			
	Lifting structure as per design	I- Beam section for pathway of trolley with fabrication and fixing.	1	LS
	Electrical chain pully with lifting system	4 tonne capacity with movement trolley Type: Electric Chain Hoist with Pendant Control	1	LS

PUMP SECTION				
7	PUMP			
	Permeate pump for drawing water from MBR membranes and sending to treated water collection tank. Backwash pump for backwashing of membranes as required. sludge transfer pump suitable for transferring sludge from aeration / MBR chamber to sludge dewatering system / storage system / drying bed. Pump should be self-priming with, working standby mode of operation, with separate isolation valve, NRV and other required accessories.			
	Feed pump	Self-priming coupled set 5 HP with 14-meter head MOC: CI, SS Impeller, MAKE: Kirloskar/CG	1 +1	NO.
	Permeate pump with stand	Self-priming coupled set 5 HP with 14-meter head MOC: CI, SS Impeller, MAKE: Kirloskar/CG	As per design	
	CIP Backwash pump with stand	Self-priming coupled set 2 HP with 14-meter head MOC: CI, SS Impeller, MAKE: Kirloskar/CG	As per design	
	RAS Pump	Sewage pump 5 HP with 14 m head MOC: CI, SS Impeller, MAKE: Kirloskar/CG	1	NO.
	Anoxic Agitator	Slow speed agitator, suitable for anoxic tank mixing, MOC: SS, Make: Kirloskar/CG	1	NO.
ELECTRIC SUPPLY SECTION				
8	PANEL			
	PLC based automatic control panel, suitable for operation of MBR with completely automatic operation as per programme given to PLC. Programme should be such that any parameters for operation of pump, blower and other equipment can be changed on site without any special support of requirement of supplier. Control panel housing should be made up of MSEP Powder coated, along with all switch gears and other components as required. Control panel should have compulsory IOT based model for sending the necessary data to a dedicated server.			
	Panel with stand	HMI AND PLC with IOT based along with VFD control Panel Housing: MSEP Powder coated	1	NO.
	Electrical wire	Electrical cable wiring, Flexible copper cable, Make: Orbit	1	LS
LIQUID PIPINGS				
10	PIPE LINE			

	Pipings	All UPVC high grade materials along with brackets and plumbing Make: Finolex/Astral	1	LS
	Flow Meter	Electromagnetic flowmeter Make: Fluidose	2	NO.
	Screen	For Pre aeration	1	NO.
	Electric Solenoid valve	For automatic valve controller Make: Aira	2	NO.
14	The scope includes the supply, transportation, loading, installation, and commissioning of all materials, along with the provision of 20 kg of bacterial culture for plant start-up and 50 litres of disinfectant chemicals for the initial trial run.			

3. DETAILED SCOPE OF WORK OF SUPPLIER

Design, engineering, supply of equipment along with accessories, piping, valves and instrumentation, all electrical items within the battery limits, erection and commissioning, testing and handing over the plant

The scope includes training of the client operators. Scope also includes first charge of oil, lubricant and chemicals.

Note:

1. The selection of equipment is based on the waste water characteristics finalised by the clients. However, if there is a variation in the characteristics or site conditions, which necessitates changes in the equipment specification or rerouting of pipelines and electrical wiring, any additional cost incurred has to be reimbursed by the client.
2. Price does not include liasoning/CFE/CFO from KSPCB and is in the scope of client.

Client's scope

1. On-site unloading, handling, lead, lift, and placement of equipment after delivery.
2. Execution of all related civil works.
3. Safe storage of equipment supplied till erection and commissioning.
4. Construction of approach roads, weather protection shed for distribution board, pump and panel.
5. Supply of utilities like electricity and fresh water of suitable quality (free from hardness, iron, and other contaminants) required for plant operation.
6. Lightning protection.
7. Ensuring proper electrical safety measures and earthing at site.
8. Provision of electrical power incomer cable/line up to the STP control panel, including all necessary protection devices and terminations.
9. Supply of all types of laboratory equipment or its test reports from any 3rd party, during & after commissioning.
10. Handling and disposal of sludge.
11. Any item or activity not specifically included in the Supplier's scope of work.

Erection and commissioning

1. Post-delivery, suppliers team requires 10 days to reach site for installation. Client is advised to complete their scope within this period to commence installation work.
2. It is the responsibility of the client to Make all the arrangements and facilities for carrying out pre-commissioning tests and commissioning test. If the client fails to Make arrangements and if

commissioning is delayed for no fault of supplier, client will release all the payments as if commissioning completed, lodging and boarding for the personnel deputed will be by the client for extended period of time.

3. If the completion of work is delayed beyond the contractual date for reasons attributed to client like delay in approval of drawing or documents or test certificate, release of clearance, lack of instructions, delay in release of payment as per the PO condition, supplier shall be entitled to suitable extension of period of completion by Mutual consent.
4. In case the plant is not commissioned within 30 days of supply due to non-availability of effluent/waste water or any other reason, the plant will be deemed to be commissioned.

4. OTHER TERMS AND CONDITIONS:

Operator training

The client will ensure that his operating personal is made available before pre-commissioning to enable them to witness all commissioning test and satisfy themselves about all guarantee parameters.

Warehouse charges

Should dispatch of equipment be delayed due to client instructions or any other reason attributable to client warehouse expenses of 0.5% of consignment value or part thereof per month, shall be payable by the supplier.

Excess material

Since spare or excess material will be supplied for uninterrupted erection and commissioning, any excess material remaining unused will be of supplier's property and will be removed from site after handing over the plant.

Warranty

All the material supplied will be covered under warranty for manufacturing defect and faulty workmanship for a period of 12 months from the date of commissioning. The warranty will not cover any and/or all the features caused due to the following reasons and shall become null and void on occurrence thereof:

- Damage or defect caused by transportation, accident, misuse, lack of maintenance, improper usage or negligence on part of the owner.
- Willful damage or negligence, normal wear and tear.
- Abuse or misuse of equipment/product.
- Damage or defect caused by alteration, modification, conversion, not authorized by supplier in writing.
- Repairs carried out by personnel other than supplier's authorized service providers.
- Defects or damages due to any other external means or causes.

Parts and components, which are repaired or replaced during such period, are warranted for the original warranty period only and will not carry any extended warranty. The replaced or defective material shall be taken back by contractor.

Force Majeure

The equipment and the total plant are subjected to standard Force Majeure clause.

5. SOME OF OUR RECENT INSTALLATIONS:



**1.6 MLD ESTAPRO
STP installed in
KARWAR CITY
MUNICIPAL CORPORATION**

**300 KLD AUSTEN PRO
ETP installed in
CAMPCO**



**AUSTENPRO
installed in
MATRIX HONDA**

**150 KLD AUSTEN PRO
ETP installed in
MEGHA FRUIT PROCESSING
VIZAG**



We look forward to your valued order